

Reg. No															
---------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Ph.D(Course Work) EXAMINATION, NOVEMBER 2025

OPEN BOOK EXAMINATION

21IPC501J - RESEARCH METHODOLOGY

Note:

- Specific approved THREE text books (Printed or photocopy) recommended for the course.
- Handwritten class notes (certified by the faculty handling the course / Head of the Department).

Time: Three hours

Max. Marks: 100

(5 × 20 = 100 Marks)

Marks BL CO

Answer any 5 Questions

- a) Identify the type of research for the following: (5 Marks)

20	4	1
----	---	---

 - The study of behavioral response of employees to the changes occurring in an organization
 - Number of mortality rate due to various reasons
 - Survey of public response before launching a product
 - Cause for the inflation in a country
 - Customers preference to buy a specific item

b) You are assigned the task to study the impact of online learning on higher education students after 2030. Design a sample questionnaire (at least 5 key questions) ensuring it meets the characteristics of a good research tool. (5 Marks)

c) A researcher develops a new biodegradable plastic material. Which form of IPR (patent, copyright, industrial design, geographical indication) is most appropriate for protecting this innovation? Explain your reasoning. Also, discuss the key statutes governing that Intellectual Property Rights (IPR) in India? (10 Marks)
- a) A researcher wants to apply a project on AI-based medical imaging for a project worth ₹75 lakhs. The government agency requires a 20% institutional contribution. (8 Marks)

20	4	2
----	---	---

 - Identify at least two potential funding agencies in India or internationally that could support this project.
 - Calculate the institutional contribution required.
 - Outline the basic structure of the project proposal.

b) List the reference for the following article in multiple citation formats. Use any four formats (4 Marks)

 - Author: Sharma, R. & Gupta, A.
 - Title: Advances in EV Charging Systems
 - Journal: International Journal of Power Electronics, Vol. 15, No. 2, pp. 101–120, 2024

c) Two journals publish articles on smart grid. One has a high impact factor; the other has many highly cited authors. Compare how journal metrics and author-level metrics can help you evaluate which article is more reliable for your research. What are the strengths and limitations of each metric, and how do they influence the journal rankings? (8 Marks)

3. a) An author publishes nearly the same study results in two different journals with slight modifications to the text and title. Is it ethical or unethical?. Justify. Also, differentiate between duplicate publication, overlapping publication, and salami slicing using this case. (10 Marks) 20 4 3
- b) A journal editor finds that a submitted manuscript has a 30% similarity index in a plagiarism check. The authors argue that most of it is from the "Methods" section and standard phrases. As an editor, how would you evaluate this case? What steps should be taken before accepting or rejecting the paper? (10 Marks)
4. a) A sample of 10 students before and after attending a coaching class obtained the following scores in an aptitude test: (14 Marks) 20 4 5
Before: 45, 50, 39, 62, 41, 55, 48, 60, 42, 58
After: 49, 55, 43, 65, 44, 59, 52, 63, 46, 61
Test whether the coaching class has significantly improved performance using paired t-test at 5% level.
- b) A researcher wants to test the fuel efficiency of electric vehicles. Standard deviation from previous studies is 3 km/kWh, and they want 95% confidence with margin of error 1 km/kWh. Calculate the required sample size. (6 Marks)
5. a) Three fertilizers are tested on crop yield. Yields (kg/plot) for 4 plots each: (10 Marks) 20 4 4
• Fertilizer A: 50, 52, 48, 51
• Fertilizer B: 55, 54, 56, 57
• Fertilizer C: 49, 47, 50, 48
Test at 5% significance whether the mean yields differ.
- b) A study records hours of study and exam scores for 10 students. (10 Marks)
• Hours: 2, 3, 5, 4, 6, 7, 8, 5, 9, 10
• Scores: 50, 55, 65, 60, 70, 72, 78, 68, 85, 90
Calculate Pearson correlation and interpret the result
6. a) Use K-means clustering ($k=3$) to group customers and describe the characteristics of each cluster. (14 Marks) 20 3 5

Customer	Age	Annual Income (\$1000)
1	25	40
2	30	60
3	22	35
4	40	80
5	36	75
6	28	50

- b) A university initiates a project on smart irrigation systems in rural areas. Identify the perspectives of at least three stakeholders and discuss how their views would shape the research design. (6 Marks) -
7. a) A random sample of 2000 people was surveyed to test whether vaccination is effective against smallpox. Out of 500 vaccinated persons, 31 were attacked by smallpox, while out of 1500 not vaccinated, 185 were attacked. Test at 5% significance level using the Chi-square test whether vaccination is effective (14 Marks) 20 4 4
- b) An automotive manufacturer records the number of defective tires produced per day. Over 30 days, an average of 2 defects/day is observed. Assuming a Poisson distribution, calculate the probability that on a given day exactly 3 defective tires are produced. (6 Marks)

* * * * *